Ariana Olivares

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Research Interests: Accessible soft robotics, low-cost manufacturing methods, and design for manufacturability.

EDUCATION

Tufts University, Medford, MA
M.S. in Mechanical Engineering, Controls and Design Track — Expected May 2027
B.S. in Mechanical Engineering, *Magna Cum Laude* — May 2026

RESEARCH EXPERIENCE

M.S. Thesis Research – Nemitz Robotics Group (Sept 2026 – Present)
Tufts University, Department of Mechanical Engineering | Advisor: Dr. Markus P. Nemitz

Injection Molding of Soft Robotic Parts with a Pellet Extruder

- **Objective:** Develop a low-cost injection molding process using a pellet extrusion head to make soft robotic parts from elastomers too soft to 3D print, toward accessible, landmine-detecting robots.
- **Contributions (in progress):** Developed a modular mold system and a shot-size calibration to diagnose fill failures; identified neck width as the limiting factor in dog-bone fills (from a short shot at 10.6 mm to a complete fill at 15 mm) and showed that optimal temperature and fill speed shift sharply between elastomers.
- **Next steps:** Benchmarking against a commercial injection molder and printed and cast parts; adding a PLA tool head so one machine prints its own molds; recording fills on camera to link process settings to outcomes.

Undergraduate Research Assistant – Nemitz Robotics Group (Sept 2024 – May 2026)
Tufts University, Department of Mechanical Engineering | Advisor: Dr. Markus P. Nemitz

Pellet Printing for Soft Robotic Devices

- **Objective:** Evaluate printing soft robots from ultra-soft elastomer pellets, bridging casting and filament printing.
- **Contributions:**
  - Led printed-versus-cast PneuNet comparison: cast silicone actuators; set up tensile and inflation-cycle testing.
  - Integrated a pellet extrusion head into a ProForge 4 tool-changing printer, adapting its firmware, G-code, and configuration to print soft and rigid materials in a single process.
  - Designed and CAD-modeled a pellet-printed soft robotic fish with internal pneumatic chambers.
- **Outcomes:** Printed actuators as soft as Shore 22A withstood 100,000+ inflation cycles without failure; co-authored a 2026 *Advanced Science* paper.

Summer Undergraduate Research Fellow (SURF) – Soft Intelligent Materials Lab (Summers 2024 – 2025)
Stanford University, Department of Mechanical Engineering | Advisor: Dr. Renee Zhao

Freestanding Magneto-Mechanical Metamaterials (Summer 2025)

- **Objective:** Build and experimentally validate metamaterials that morph from flat sheets into wave-shaped 3D surfaces, collaborating with Northwestern University, which simulated the multi-wave designs.
- **Contributions:** Developed a fabrication and magnetization process for silicone–NdFeB structures to match simulations, designing fixtures to set each row's magnetization angle (−45° to 45°); built a MATLAB program that measures 3D deformation from two photos to compare against simulations; simulated the single-wave design in ABAQUS and MATLAB, matching measured depth within 14%, with gaps traced to friction and tolerances.

Conformal Magneto-Mechanical Metamaterials (Summer 2024)

- **Objective:** Design magnetic metamaterials that conform to and actuate on cylindrical, concave, and convex surfaces.
- **Contributions:** Designed unit cells with tailored widths and side angles, then molded and magnetized 100+ silicone–NdFeB cells; actuated 12×5 arrays at 50 mT and a 16×3-cell cylinder with an iron-core coil, finding that raising magnetic particle loading from 25% to 35% enabled even deployment, validated in ABAQUS.

PUBLICATIONS

Wu, Y., Chen, J.-H., **Olivares, A.**, Kostak, K., Pedicone, S., Kendre, S. V., Nemitz, M. P. *Pellet Printing for Soft Robotic Devices. Advanced Science*, 13(30), e24237 (2026). https://doi.org/10.1002/advs.202524237

POSTER PRESENTATIONS

Olivares, A., Sim, J., Hwang, S., Zhao, R. R. *Freestanding Magneto-Mechanical Metamaterials.* Stanford SURF Symposium, Stanford University, August 2025.
Olivares, A., Wu, Y., Nemitz, M. P. *Pellet Printing for Soft Devices.* Mechanical Engineering Research Symposium, Tufts University, Spring 2025.
Olivares, A., Sim, J., Hwang, S., Zhao, R. R. *Design and Actuation of Conformal Magneto-Mechanical Metamaterials.* Stanford SURF Symposium, Stanford University, August 2024.

TEACHING EXPERIENCE

Teaching Assistant – Senior Capstone Design (Sept 2026 – Present)
Tufts University, Department of Mechanical Engineering | Instructor: Dr. Briana Bouchard

- Advise 40 students through the full design cycle, from requirements and concepts to final design reviews.
- Give technical feedback on CAD models, engineering drawings, analyses, and fabrication plans.

Learning Assistant – Engineering Design (Jan 2025 – May 2025)
Tufts University, Department of Mechanical Engineering | Instructor: Dr. Kristen Wendell

- Mentored 40+ students in SolidWorks, engineering drawings, and prototyping for user-centered design projects.

Learning Assistant – Materials and Manufacturing (Sept 2024 – Dec 2024)
Tufts University, Department of Mechanical Engineering | Instructor: Dr. Luisa Chiesa

- Guided first-year students through CNC, laser cutting, and 3D printing, supervising 15+ final projects.

HONORS AND AWARDS

- Honos Civicus Society Inductee (2026) — Tisch College of Civic Life honor for academics and civic leadership
- Tufts Engineering Excellence Prize (2025) — Department award for academic achievement and research
- Stanford Engineering SURF Fellow (2024, 2025) — Selective national summer research program
- SHPE Linde Engineers of Tomorrow Scholar (2025) — National award for leadership and service
- SHPE Undergraduate Scholarship (2024) — National merit-based scholarship
- YMCA Latino Achiever Scholarship (2022–2025) — Academic excellence and community engagement

LEADERSHIP AND OUTREACH

Breakthrough San Francisco – Teaching Fellow (June 2026 – Aug 2026)

- Taught 7th grade math, designing lessons on financial literacy, proportional reasoning, and problem-solving.
- Led a daily advisory of students, mentoring them through goal setting, relationship building, and academic support.

Tufts Center for Engineering Education and Outreach – Learning Fellow (May 2023 – Present)

- Teach engineering design, robotics, and programming to 100+ K–8 students through hands-on activities.
- Develop weekly lesson plans and STEM modules tailored to diverse learning levels.

Society of Hispanic Professional Engineers (SHPE), Tufts Chapter – Corporate Liaison (Sept 2022 – Present)

- Organized résumé workshops, mock interviews, and corporate info sessions for engineering students.
- Built partnerships with Google, Microsoft, Vertex, MITRE, GE Aerospace, and Bloomberg.

TECHNICAL SKILLS

Programming & Robotics: Python, MATLAB, C++, G-code, ROS2, Raspberry Pi, MicroPython
Modeling & Simulation: ABAQUS, SolidWorks (certified), Fusion 360, Onshape | **Languages:** English, Spanish (fluent)
Fabrication: CNC, Injection Molding, Mold Design, Laser/Water-Jet Cutting, 3D Printing (SLA, FDM, pellet)
Materials & Testing: Silicone Casting, Magnetization, UTM and Cyclic Testing, Image-Based Deformation Measurement